Problem Chosen	2024	Team Control Number
D	**MCM/ICM Summary Sheet**	**2408456**

Modeling of water level control in the Great Lakes

Summary

In order to better maintain the stability of the water level of the Great Lakes and reduce the disasters arising from the polarization of the water carrying capacity of the rivers and lakes, this study will be based on the historical water level data of the Great Lakes Basin, to establish a mathematical model of the water level management mechanism that can be adjusted, in order to be more committed to the development of the local aspects.

For problem 1: Through the Great Lakes water level data given in the question, within the error interval of 0.9m, using the monthly average data of water level in previous years, reflecting the change of water level in a year through the **fluctuation rate**, and selecting the fluctuation rate **in the tolerable range** of the data as the optimal water level interval in the month, and finally obtaining **the optimal water level** in each month of the Great Lakes, respectively, as shown in **chart 2**.

For problem 2: the Great Lakes water system is intricate, the interaction between the rivers and lakes will lead to changes in their water levels. In solving this problem, we determine the magnitude of the change in water level by constructing **a differential equation for the lake system** with **small changes in the variables**. Substituting the data of lake inflows and outflows, we finally obtained the solution to maintain the optimal water level of the Great Lakes as shown in **figure 8**.

For Problem 3: We first found **the relationship function between water level change and dam height**, and obtained the sensitivity of the Great Lakes water level to dam regulation by solving the **derivative function**. When quoting the 2017 water level data, we determine whether the water level is optimized or not by calculating the **Euclidean distance** before and after the change in water surface height, and finally we get the results shown in the **figure 9**.

For Problem 4: This question is to investigate whether the above algorithm is sensitive to natural phenomena, and this question takes **the root mean square error RMSE of the water level control performance** as a sensitivity index, and establishes the relationship between the three types of natural factors and the root mean square error. **The coefficients of ice blockage** occurring in Lake Superior, Lake Michigan, and Lake Erie in the spring are **0**, the coefficients of ice blockage in Lake Huron and Lake Ontario are **1,** and **the sensitivity coefficients** of precipitation, winter snowpack, and ice blockage are **4.57**, **2.36**, and **-3.47**, respectively.

For Problem 5: This problem is based on the requirements of six types of stakeholders for Lake Ontario,we optimize and modify the model by **fitting the water quantity function** and using its **derivative function** as **an indicator of the rate of change of the water level**, and obtaining the target water level change that can relatively satisfy the stakeholders through **AHP**, and optimizing and modifying the model of the parameter of the change of the water level.The optimized water level function is obtained as in **figure 11**.

Keywords: Differential Equation Modeling, RMSE,AHP,Fitting Algorithm

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1 Introduction

1.1 Background

As the largest group of freshwater lakes in the world, the Great Lakes of North America carry 20% of the world's freshwater resources and also play a pivotal role in many aspects [1]. The Great Lakes cover a wide area, and the climatic conditions of the cities through which it passes also differ greatly. Because of this, the Great Lakes basin as a system of upstream and downstream connected lakes forms a dynamic network flow, in which rivers and lakes are interdependent and interact with each other with complex requirements and uncertainties that make this network flow more difficult. If the processes of lake drainage and evaporation consume too little water, they can lead to flooding, which will have a direct impact on local residents and businesses, and if they drain too much water, they are impassable to large vessels, which will have a direct impact on the local economy [2]. It is imperative to find the optimal water level of the Great Lakes to better solve these problems. This question is a study based on various climatic and hydrological information of the Great Lakes, with the geographic profile of the North American Great Lakes as a background, and the impact of the Great Lakes water level management system on the desires of various industry stakeholders.

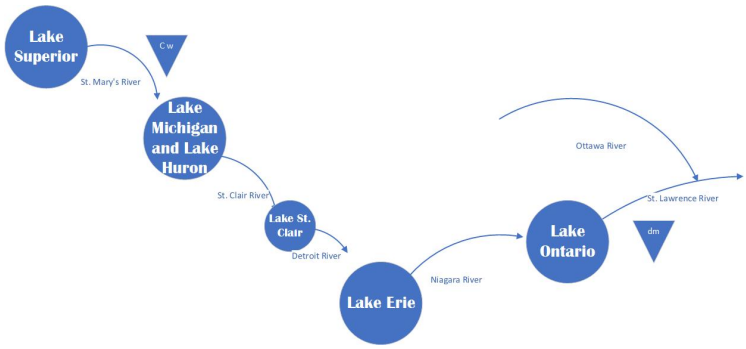

Figure 1:Great Lakes Simulation

1.2 Restatement of problems

Because the complexity of the Great Lakes flow network is difficult to deal with in a simple manner, the IJC needs ICM to construct a control mechanism for water levels in the basin to assist in their management of the Great Lakes.

The specific tasks we need to accomplish are listed below:

Problem 1: Under the premise of meeting the desires of the various stakeholders of the Great Lakes, the optimal water level for the Great Lakes needs to be identified for each month of the year. That is to say, the selection of this water level should not only satisfy the interests of each stakeholder as much as possible, but also take into account the Great Lakes as an upstream and downstream connectivity of the lake system, the combination of these factors to make the Great Lakes reach the optimal level, that is, the optimal water level we require.

Problem 2: Lake Superior, Lake Michigan and Lake Huron, Lake Erie, and Lake Ontario are connected by rivers to form a complex hydrological system, which leads to dynamic changes in water levels and influences each other, and the water levels of the Great Lakes are related to many stakeholders. Question two requires the establishment of a model that can simulate the inflow and outflow of water from each lake of the Great Lakes, so that the water level of each lake of the Great Lakes can be

obtained to adjust the parameters of the model to achieve the level of the Great Lakes can be maintained at the optimal water level.

Problem 3: In the previous question, question 2, we have initially established the water level regulation and control model of the Great Lakes, question 3 is established on the basis of the model established in question 2, and it is required to evaluate the sensitivity of the water level algorithm model established in question 2, and to analyze its sensitivity regarding the heights of the two dams. Then, using relevant data from 2017, the model was used to make predictions about water levels and determine whether changes in water levels under the algorithm would make make make stakeholders more satisfied.

Problem 4: The environment tends to have an impact on water level changes in lakes, and although human interventions such as regulating dams can make lake levels tend to stabilize, the reality is that environmental factors such as precipitation, ice jams, evaporation, and other phenomena that cannot be intervened with by humans still affect significant water level changes. Problem four requires analyzing the sensitivity of the control algorithms we have developed to environmental changes, and then reflecting the level of stability of the control algorithms in controlling water level changes.

Problem 5: Lake Ontario in the Great Lakes in many ways with the other four lakes there are large differences, such as Lake Ontario's lake area is small, and is in the Great Lakes downstream, the year's water level changes are very large (Figure), and Lake Ontario is in the direct and the Atlantic Ocean convergence of the first lake, the value of shipping is very high, and current residents of Yan'an is very concerned about the water level of the lake and many other differences in the factors, problem Fifth is to combine the many factors related to water level in Lake Ontario to expand our model above to analyze for Lake Ontario to establish a more appropriate and comprehensive model, on the basis of the above considerations to increase the consideration of the factors.

1.3 Analysis of problems

For Question 1: The first question has two main core stakeholders, which are the residential and commercial facilities along the lakeshore and the local economy, while the flora and fauna are secondary stakeholders in comparison. We have obtained the past water level data of the Great Lakes from the Annex, and by processing the monthly water level data of the Great Lakes for each year given in the Annex to obtain the annual average water level, and then calculating the rate of change of water level (the ratio of the difference between the water level change and two feet) and the standard deviation of the fluctuation of the water level of each lake to determine the optimal water level of the Great Lakes through the above two indexes.

For problem two: the Great Lakes water level changes are mainly due to changes in the amount of water in each of the lakes of the Great Lakes, and we believe that changes in the amount of water that occurs in the way of the main two categories, one is the net inflow of nature, which we refer to as the natural net inflow, including a number of human-induced uncontrollable factors, there are rainfall, snow and ice melt, evaporation, infiltration, but in this problem we will be the net inflow of natural water in each lake But in this problem, we regard the natural net inflow of each lake as a constant quantity, and do not discuss it in detail for the time being; another type of way to increase the amount of water, but also the most important way is the flow of water, the water volume of a lake is mainly determined by the inflow and outflow of water, and the outflow of water can be determined by the inflow of lakes downstream of the lake, so there is a correlation relationship between the two, but also because we are mainly concerned about the change in the amount of water of the lakes, and therefore consider the construction of the lake system Modeling of differential

equations.

For Problem 3: This problem is divided into two sub-problems, one is to firstly analyze the sensitivity of the model established in Problem 2 for the water level change regarding the two dams, and the other is to actually substitute the corresponding data of the Great Lakes in 2017, and then simulate the adjustment of the height of the two dams, and then analyze the result of the water level change to analyze whether it makes the stakeholders more satisfied. For subproblem one, we will find the relationship between water level changes and dam heights based on the model in problem two, and analyze the sensitivity of water level changes in each lake with respect to the sum of the two dam heights by solving for the skewness in the function. For sub-problem two, first we will collect the data of the Great Lakes in 2017, substitute them into the model for calculation and simulation, and then get the original water surface height of the Great Lakes and the water surface height after the change of the data, and then use the Euclidean distance to calculate the distance of the water level data before and after from the optimal water level solved in Problem One, and observe whether the distance is increasing or decreasing in order to judge whether it makes the stakeholders more satisfied or not. more satisfied.

For Problem 4: In Problem 2, we have established the optimal water level control model for the Great Lakes, according to the requirements of the topic, we need to explore the sensitivity of the environmental factors on the water level of the Great Lakes, for the environmental factors we mainly consider the precipitation of the Great Lakes, the ice jamming and the evaporation of the lake water. We choose the root-mean-square error of the water level control performance as the sensitivity index to establish the relationship between the system performance and the environmental condition variables.

For Problem 5: By reviewing the information and reading the known conditions in the question, it is found that there is a sub-problem of Lake Ontario, i.e., the conflict of interest problem of the water level of Lake Ontario, so the extensive analysis of the model based on the real conditions of Lake Ontario needs to take into account three factors, one is the major stakeholders, and the other is the other factors that can determine the water level of Lake Ontario. First we will analyze and fit the existing data on Lake Ontario water levels using least squares to fit the natural water level changes in Lake Ontario over the course of a year to find the existing water level changes where there is a greater concern. Then for the stakeholder analysis, we mainly consider six stakeholders that are directly related to the water level of Lake Ontario, which are residents of Lake Ontario, residents of Montreal ports, shipping companies, fishing boats, environmentalists, power generation companies, who have different water level needs, we use hierarchical analysis to find out the corresponding weights of these stakeholders, and based on the weights, we use a hierarchical analysis to find out the corresponding weights of the above model. Based on the weights, we optimize the above model in terms of the maximum value and volatility, and then weight the respective interests to obtain a Lake Ontario water level model that can take into account the interests of all parties as much as possible.

1.4 Overview of our work

This problem actually requires us to model the water level of the lake system of the Great Lakes and to develop reasonable strategies to satisfy the needs of different stakeholders for water level. To avoid complex descriptions and to visualize our workflow, the flowchart is shown in Figure:2.

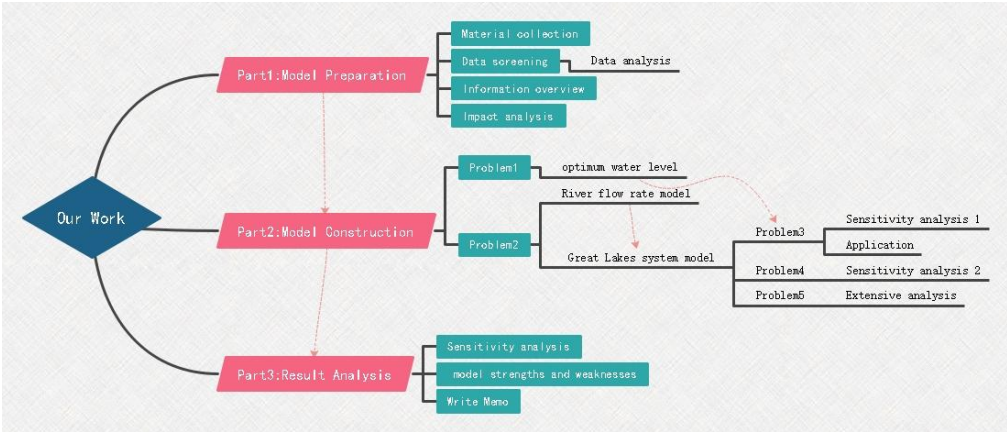

Figure:2 Overview of our work

2 General Assumptions and Variable Description

2.1 General Assumptions

Assumption 1: Lake Michigan and Lake Huron are analyzed as a whole. Lake Michigan and Lake Huron are directly connected, which is equivalent to the existence of a river connection between the two with a positive infinite flow rate, so that the two have equal liquid surface heights at any given moment can be analyzed as a whole by using the idea of the holistic approach.

Assumption 2: Neglecting the amount of water in Lake St. Clair, the St. Clair River, Lake St. Clair, and Detroit River are equivalent to a river whose flow is the lesser of the flow of the St. Clair River and the flow of the Detroit River at this point in time. We find that Lake St. Clair, a small lake located between Lake Erie and Lake Huron, has a very small volume of water relative to either of the Great Lakes, and therefore we consider it to have no storage capacity, no absorptive or augmentative effect on the flow of water in the Great Lakes, and that Lake St. Clair and Lake Huron are connected by the St. Clair River, while the Detroit River is connected by the St. Clair River, which is the smaller of the St. Clair River flow and the Detroit River flow. Clair River and Lake Erie through the Detroit River, we can equate the St. Clair River, Lake St. Clair, and the Detroit River to a river that connects Lake Erie and Lake Huron directly. The river flow is the smaller of the St. Clair River flow or the Detroit River flow at that point in time:

$$Q_{i=2} = \min\left\{Q_{St.\ Clair\ River}, Q_{Detroit\ River}\right\}$$

Assumption 3:Climatic conditions in the Great Lakes basin are considered stable [3] and do not change significantly in the model we calculated.

Assumption 4:Depreciation of rivers due to other factors during flow is not considered.

2.2Variable Description

Symbols	Meanings	Dimension
h	Water level	m
H	Optimal water level range	m
θ	Monthly fluctuation rate of water level	
V	Volume of the lake	m^3

t	Time	month
N	The natural increase in the amount of water in a lake	m per month
α	River flow rate	
$h_{\max}$	The maximum bearing water level of a lake	m
S	The surface area of the lake	m^2
Q	Discharge of river	m^3/s
SF	The basin area of the lake	m^2
M	The discharge from the upper lake to the lower lake	m^3
h'	The water level of the lake after the change	m
d	Ideal deviation of liquid level	m
Vn	The natural inflow of the lake	m^3
T	Mean air temperature	℃
$rain$	Total lake precipitation	m^3
$evoporation$	Evaporation of the lake surface	m^3
w	Weight coefficient	

Note: Meaning of subscript

‘j’ stands for four lakes, from 1 to 4: LakeSuperior, LakeMichiganandLakeHuron, LakeErie and LakeOntario.

‘i’ on behalf of the three rivers, from 1 to 3 are: st. Mary 'sRiver, st. ClairRiverandLakeSt. ClairandDetroitRiver, NiagaraRiver

‘s’ stands for four seasons, from 1 to 4: Spring, summer, autumn and winter

‘m’ stands for 12 months of the year.

‘p’ represents the stakeholders, from 1 to 6 representing the residents of Lake Ontario, the residents of the Port of Montreal, shipping companies, fishing boats, environmentalists, and power generation companies.

3 Model Establishment and Solutions

3.1 Problem1

3.1.1 Model Establishment

In this case, we take the month as the time dimension, and record the monthly water level data of the Great Lakes in 23 years as $h_{j,y,m}$, and then obtain the average water level of the Great Lakes $h_{j,m}$ in the same month in 23 years:

$$\overline{h_{j,m}} = \sum_{y=2000}^{2022} h_{j,m,y} / 23$$

After the average water level of each month in 23 years is obtained, the average

value of the above data $\overline{h_j}$ is calculated to obtain the average annual water level of the Great Lakes, where (j is 1,2,3,4; y let's take 2000,2001..... 2022; m take 1,2,3..... 12) :

$$\overline{h_j} = \sum_{m=1}^{12} \overline{h_{j,m}}$$

Due to the short time span of 23 years, we believe that the climate and environmental conditions of the Great Lakes basin did not change significantly during this period, so the water level of the lakes always conforms to the seasonal changes, and the annual change rate of each lake is calculated λ_j :

$$\lambda_j = (\overline{h_{j,m}} - \overline{h_j}) / \overline{h_j}$$

According to the title, we know that a normal change in water level of 2 to 3 feet will have a significant impact on some stakeholders. Therefore, we believe that the more smoothly the water level of the lake changes, the more beneficial the more stakeholders, and the results are more consistent with the optimal water level. We stipulate that the maximum deviation between the real water level and the calculated mean water level is $\Delta_{\max} = 2$ feet, 1 foot =0.3048m, and this difference $\Delta_{\max}$ is compared with the annual comprehensive average water level of each lake to obtain the water level fluctuation rate θ :

$$\theta_j = \Delta_{\max} / \overline{h_j}$$

The optimal water level interval is determined by the floating rate of water level, and the optimal water level of the Great Lakes is finally determined:

If for a month, there are:

$$|\lambda_j| <= \theta_j$$

The best water level for the month $H_{j,m}$ is:

$$H_{j,m} = \overline{h_{j,m}}$$

If $|\lambda_j| > \theta_j$:

$$H_{j,m} = \overline{h_j} \bullet (1+\theta_j), \overline{h_{j,m}} > \overline{h_j}$$
$$H_{j,m} = \overline{h_j} \bullet (1-\theta_j), \overline{h_{j,m}} < \overline{h_j}$$

3.1.2 Model Solving and Analysis of results

Based on the above analysis, we can obtain the following annual water level values for the Great Lakes:

Chart 1:Annual water level values

lake	average
Lake Superior	183.34
Lake Michigan and Lake Huron	176.33
Lake Erie	174.28
Lake Ontario	74.83

Chart 2:Monthly water level values

Month&name	Lake Superior	Lake Michigan and Lake Huron	Lake Erie	Lake Ontario
Jan	183.27	176.20	174.15	74.66
Feb	183.21	176.88	174.14	74.72
Mar	183.17	176.20	174.23	74.76
Apr	183.21	176.28	174.35	74.94
May	183.31	176.38	174.43	75.11
June	183.40	176.46	174.48	75.15
July	183.46	176.49	174.46	75.10
Aug	183.47	176.47	174.39	74.97
Sep	183.46	176.41	174.29	74.78
Oct	183.45	176.34	174.19	74.63
Nov	183.41	176.29	174.13	74.56
Dec	183.35	176.25	174.13	74.57

The deviation value of water level fluctuation is: Lake Superior 0.0033, Lake Huron & Lake Michigan 0.0035, Lake Erie 0.0035, and Lake Ontario 0.0081.

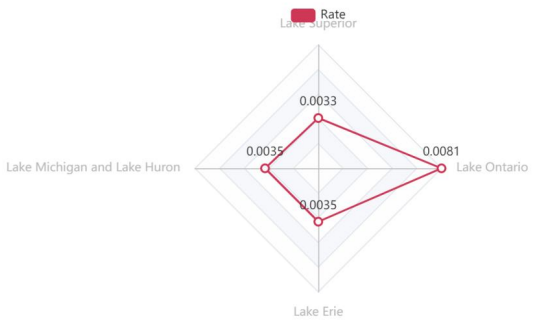

Figure 3:The deviation value of water level fluctuation

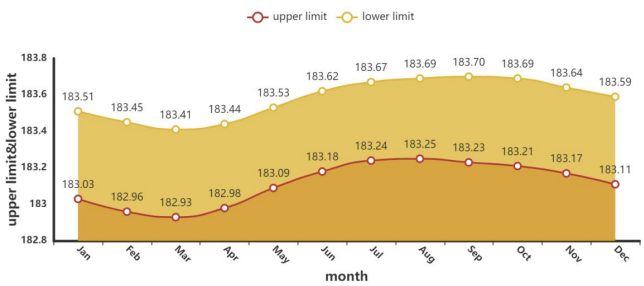

Figure 4:Lake Superior's upper limit and lower limit of water level

We can find that the deviation value of Lake Ontario is larger, while the deviation value of the other three parts of the lake is smaller. The average water level is enveloped based on the optimal water level interval of deviation value. Combined

with the existing data, it can be seen that the original data was formed based on the vast majority of stakeholders, so the water level of the Great Lakes is always within the optimal water level range, which has met the needs of stakeholders. Therefore, the final optimal water level value is the comprehensive annual average water level obtained above.

3.2 Problem2

3.2.1 Model Establishment

Assumption 1: For lakes of the Great Lakes, the change in volume of water ΔV is approximately equal to the change in surface area of the lake S_j multiplied by the change in water level Δh. Based on the facts that have occurred, we find that all the events occurring in a lake have very little impact on the water level, and it is difficult to exceed the thousandth of the total water level in the most cases. Therefore, it is considered that the change of water level is very small compared with the change of the whole lake water body. Therefore, I believe that the area of water body has not changed within the range of water level change. As a regular geometry, its volume:

$$\Delta V = S_j \cdot \Delta h$$

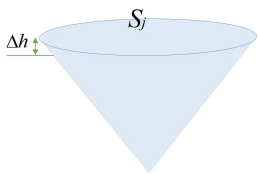

Figure 5:Assumption 1

Assumption 2: When a river is at its maximum annual flow, it is considered to be fully flowing, and frozen water does not flow at all. It is assumed that when the river level is at its maximum, the water flowing from the upstream lake can flow completely and quickly through the river to the next lake. This process takes almost no time, as long as the upstream lake leaks water to the downstream lake, it can flow completely and quickly to the downstream lake. When the river freezes, there is no carrying force on the river, and the water leaking upstream cannot reach the downstream.

Assumption3: Natural and hydrological conditions are stable. Through consulting and analyzing the meteorological data of the Great Lakes over the years, we find that the climate conditions in the Great Lakes region are relatively stable from year to year. Therefore, extreme weather and large changes in natural conditions will no longer be considered in our model, so each parameter is relatively fixed constant from year to year. The Great Lakes region is a temperate continental climate zone, but these parameters change with the months and seasons of the year, and these changes are relatively fixed from year to year.

Assumption 4: The basic unit of time t of this model is that the characteristics of rivers and lakes and the climate in a month hardly change greatly in a month, and it is

reasonable to take the average value of the month as the characteristic value of the month.

Assumption 5: Determination of the maximum bearing water level of natural lakes. For a lake with a dam, the maximum bearing water level is the height of the dam. For a lake without a dam, it should theoretically be the lowest point of the riverbed. For the sake of simplification, we assume the riverbed height at the junction with the downstream lake.

Assumption 6:We use sea level as the zero level plane with height 0. To facilitate uniform comparison, we use sea level as the zero level plane.

Model 1: River flow rate model

We take into account that there is an efficiency problem in the transport process of water from upstream to downstream. In summer, water flows at a fast rate with a large flow rate, while in winter, the flow rate is small and the flow rate is slow. In addition, the Five Great Lakes are located in the temperate continental climate zone, and the phenomenon of river icing may occur, so the transport efficiency is 0. We believe that when a river is at its maximum discharge in a year, the water flow is sufficient, that is, the water transport efficiency in summer is infinite, so it can be considered that the water output from the upstream lake can be completely input into the downstream lake in summer.

Therefore, the river flow rate α should be mainly related to the flow of the river, and the two are positively correlated:

$$\alpha \propto Q$$

We use river flow data to quantify river flow rates α :
For rivers i , there is flow data $Q_{i,m}$ for each season s of the year:
For a given river j, according to assumption 2 above, there are:

$$\alpha=1, s.t. Q_{i,m} = \max\left\{Q_{i,m}\right\}, m = 1, 2 \cdots, 12$$

The amount of water transported by a river is V, the flow is Q, and the time is t:

$$V = Q \bullet t$$

The amount of water transported in a river can also be determined by the amount of water drained upstream M and α :

$$V = M \bullet \alpha$$

So:

$$\alpha = \frac{t}{M} Q$$

It is found that the flow coefficient α is directly proportional to the discharge Q, and because the maximum discharge is defined previously α=1, the following formula can be constructed (for the same river i) :

$$\alpha = \frac{Q_m}{\max\{Q_m\}}, m = 1, 2, \cdots, 12$$

$$\alpha \in (0,1]$$

Meeting the assumptions and definitions, the parameter $\alpha_{i,m}$ can be taken as a reasonable flow rate of the river i in the month m.

Model 2: Differential equation model of water quantity variation in the Great Lakes system

The Great Lakes are a system of interconnected lake groups. One lake is selected first for analysis. The water quantity of lakes can be divided into two main categories. First, the natural net inflow, which is recorded as the natural net inflow per unit basin area, contains many factors that are not controllable by humans, such as rainfall, ice and snow meltwater, evaporation, and infiltration [4]. In this case, we regard the natural net inflow of each lake as a constant.

The other way in which water can be produced, and the main way in which it can be produced, is the flow of water. The amount of water in a lake is mainly determined by the amount of water flowing in and out of the lake, and the amount of water flowing out of the lake can determine the inflow of the lake downstream.

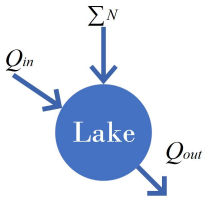

Figure 6: The water quantity of lakes

The amount of water changes in a given lake ΔV is therefore determined by natural variability $\sum N$, inflow Q_{in} and outflow Q_{out}:

$$\Delta V = \sum N + Q_{in} - Q_{out}$$

For $\sum N$,The natural net inflow per unit basin area N is multiplied by the basin area SF, which is equivalent to the area of a lake that can collect water:

$$\sum N = N \bullet SF$$

For inflow Q_{in} , it is equal to the outflow of the upstream river Q_{inj-1} multiplied by the flow coefficient of the river connected to the upstream α_{j-1}:

$$Q_{inj} = \alpha_{j\text{-}1} \bullet Q_{outj\text{-}1}$$

For Q_{outj-1} , the difference of the existing height minus the maximum carrying

height multiplied by the water surface area is taken as the outflow water. According to the assumption: for a lake with a dam, the highest carrying water level is the height of the dam; For a lake without a dam, is the river bed height at the junction with the downstream lake, so for the highest bearing water level, the artificially controllable height is $h_{1\max}$ and $h_{4\max}$, mainly by controlling the dam, and is a constant determined by nature:

$$Q_{out_{j-1}} = (h_{j-1} - h_{j-1\max}) \cdot S_{j-1}$$

Let $j = j+1$:

$$Q_{out_j} = (h_j - h_{j\max}) \cdot S_j$$

In summary, we can obtain:

$$\Delta V_j = N_j \cdot SF_j + \alpha_{j-1}(h_{j-1} - h_{j-1\max}) \cdot S_{j-1} - \alpha_j (h_j - h_{j\max}) \bullet S_j$$

In the assumption that the smallest unit of time is month, differentiate t by V:

$$\Delta V_j = \frac{dV_j}{dt}$$

plugging $j = 1,2,3,4,5$ into the models in turn:

$$\begin{cases} \dfrac{dV_1}{dt} = N_1 \cdot SF_1 - \alpha_1 (h_1 - h_{1\max}) \cdot S_1 = \Delta V_1 \\ \dfrac{dV_2}{dt} = N_2 \cdot SF_2 + \alpha_1 (h_1 - h_{1\max}) \cdot S_1 - \alpha_2 (h_2 - h_{2\max}) \bullet S_2 = \Delta V_2 \\ \dfrac{dV_3}{dt} = N_3 \cdot SF_3 + \alpha_2 (h_2 - h_{2\max}) \bullet S_2 - \alpha_3 (h_3 - h_{3\max}) \bullet S_3 = \Delta V_3 \\ \dfrac{dV_4}{dt} = N_4 \cdot SF_4 + \alpha_3 (h_3 - h_{3\max}) \bullet S_3 - \alpha_4 (h_4 - h_{4\max}) \bullet S_4 = \Delta V_4 \end{cases}$$

Calculate water level change:

$$\Delta h = \Delta V_j / S_j$$

Changed water level h_j':

$$h_j' = h_j + \Delta h_j$$

Determine whether h_j' is within the optimal water level interval solved by the first question:

If:

$$h_j' \in H_j$$

It indicates that the water level is in the optimal range and does not need to be adjusted.

If :

$$h_j' \notin H_j$$

The water level is not in the optimal range at this time, and human intervention is required to adjust the controllable parameters($h_{1\max}$ and $h_{4\max}$) at this time,It is then substituted again as the new initial water level height:

$$h_j' \rightarrow h_j$$

Iteratively find a set of feasible solutions ($h_{1\max}$, $h_{4\max}$) so that the water level can return to the optimal water level interval. If it does not exist, the relevant personnel shall be notified in time to take corresponding measures.

3.2.2 Model Solving and Analysis of Results

Obviously, the core model of this problem is Model 2: differential equation model of changes in water quantity of the Great Lakes lake system. However, the premise of solving model 2 is to obtain an important parameter: river flow rate α , so it is necessary to solve model 1 first.

The steps to solve Model 1 are as follows:

Data preparation: Obtain the flow data of each river for each month of the calendar year;

Data preprocessing: Due to the small amount of data changes and the stability of the data (see the attached figure), we used the average value of the month in the past year as the substitute value for the missing data. No other outliers were found. The river flow data of St.ClairRiver and DetroitRiver are obtained for each year and month with a smaller value as $i=2$ river flow data.

Data calculation: Calculate the average flow rate of each river by month, and then find the maximum flow rate of each river in 12 months, and then calculate the flow rate α of each river in each month。

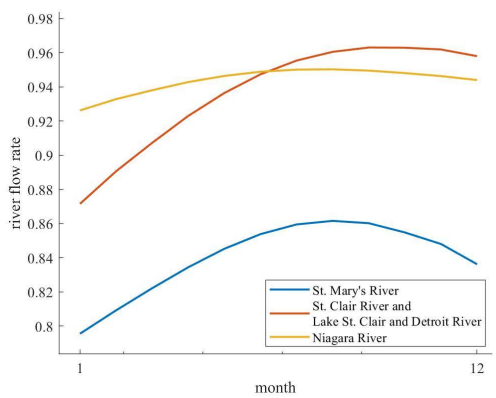

Figure 7:River flow rate

The specific steps to solve model 2 are as follows:

The parameters of the model were determined, and the natural net inflow per unit basin area N in the Great Lakes region was obtained by consulting the corresponding meteorological and hydraulic data. The basin area SF_j and lake area S_j of each lake of the Great Lakes can be obtained by consulting the data, the optimal water level of each month obtained from question 1, the dam height of compensationwork $h_{1\max}$ and MosesDam $h_{4\max}$ in each season can be obtained by consulting the water conservancy literature,in nature $h_{2\max}$ and $h_{3\max}$ can be obtained by consulting the data.

Data preparation: Obtain monthly water level data for each of the Great Lakes from 2000 to 2022.

Data preprocessing: No missing values or outliers were found.

Data calculation: The initial water level data h_j of five lakes with 12 months of data were substituted into the model for solving to get h_j'.

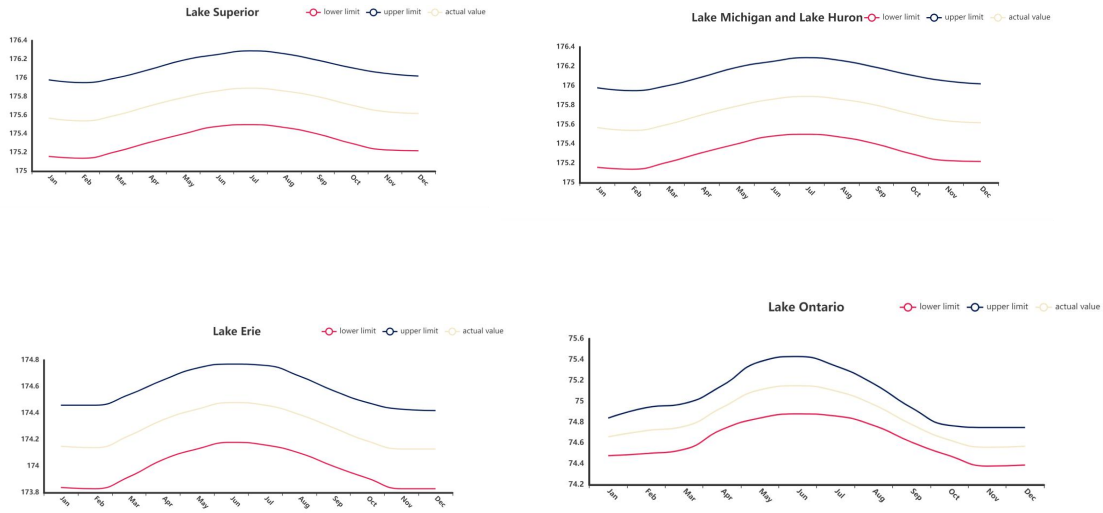

Figure 8：actual value and limit

3.3 Problem3

3.3.1 Model Establishment

Sensitivity analysis model for dams:

According to the model in question 2:

$$\begin{cases} \Delta V_1 = N_1 \cdot SF_1 - \alpha_1 (h_1 - h_{1\max}) \cdot S_1 \\ \Delta V_2 = N_2 \cdot SF_2 + \alpha_1 (h_1 - h_{1\max}) \cdot S_1 - \alpha_2 (h_2 - h_{2\max}) \cdot S_2 \\ \Delta V_3 = N_3 \cdot SF_3 + \alpha_2 (h_2 - h_{2\max}) \cdot S_2 - \alpha_3 (h_3 - h_{3\max}) \cdot S_3 \\ \Delta V_4 = N_4 \cdot SF_4 + \alpha_3 (h_3 - h_{3\max}) \cdot S_3 - \alpha_4 (h_4 - h_{4\max}) \cdot S_4 \end{cases}$$

$$\Delta h = \Delta V_j / S_j$$

So:

$$\begin{cases} \Delta h_1 = N_1 \cdot \dfrac{SF_1}{S_1} - \alpha_1 (h_1 - h_{1\max}) \\ \Delta h_2 = N_2 \cdot \dfrac{SF_2}{S_2} + \alpha_1 (h_1 - h_{1\max}) \cdot \dfrac{S_1}{S_2} - \alpha_2 (h_2 - h_{2\max}) \\ \Delta h_3 = N_3 \cdot \dfrac{SF_3}{S_3} + \alpha_2 (h_2 - h_{2\max}) \cdot \dfrac{S_2}{S_3} - \alpha_3 (h_3 - h_{3\max}) \\ \Delta h_4 = N_4 \cdot \dfrac{SF_4}{S_4} + \alpha_3 (h_3 - h_{3\max}) \cdot \dfrac{S_3}{S_4} - \alpha_4 (h_4 - h_{4\max}) \end{cases}$$

Take the partial derivative of each Δh_j with respect to $h_{1\max}$ and $h_{4\max}$. As it is a sensitivity analysis, take the absolute value of the result and get:

$$\begin{cases} \left|\dfrac{\partial \Delta h_1}{\partial h_{1\max}}\right| = \alpha_1, \left|\dfrac{\partial \Delta h_1}{\partial h_{4\max}}\right| = 0 \\ \left|\dfrac{\partial \Delta h_2}{\partial h_{1\max}}\right| = \dfrac{S_1}{S_2}\alpha_1, \left|\dfrac{\partial \Delta h_2}{\partial h_{4\max}}\right| = 0 \\ \left|\dfrac{\partial \Delta h_3}{\partial h_{1\max}}\right| = 0, \left|\dfrac{\partial \Delta h_3}{\partial h_{4\max}}\right| = 0 \\ \left|\dfrac{\partial \Delta h_4}{\partial h_{1\max}}\right| = 0, \left|\dfrac{\partial \Delta h_4}{\partial h_{4\max}}\right| = \alpha_4 \end{cases}$$

Satisfaction model:

First, we choose the optimal water surface height H_j solved in problem 1 as an ideal liquid surface height.

Since it is stated in the appendix that IJC hopes to control the water level by the dam to stabilize it near the annual mean value, we can think that the best water level value solved in the first question is approximately equal to the annual mean value, and this index is relatively natural with little human intervention, and this index is better than the annual mean value.

Therefore, a Euclidean distance formula is established, d is recorded as the ideal deviation value of the liquid level to quantify the distance from the optimal liquid level:

$$d = \sqrt{\sum_{j=1}^{4} (h_j - H_j)^2}$$

The size of d can directly reflect the deviation between the actual water surface height h_j and the optimal water surface height H_j.

Therefore, it is possible to compare the size change of d to conduct a comparative analysis of stakeholder satisfaction, and the ideal deviation value of liquid level before and after the change in a certain month d and d':

$$d = \sqrt{\sum_{j=1}^{4} (h_j - H_j)^2}$$

$$d' = \sqrt{\sum_{j=1}^{4} (h_j' - H_j)^2}$$

If :

$$d' < d$$

It shows that the overall satisfaction has increased.

If:

$$d' > d$$

It shows that the overall satisfaction has not been increased.

3.3.2 Model Solving and Analysis of Results

Sensitivity model solution:

$$
\begin{cases}
\left|\dfrac{\partial \Delta h_1}{\partial h_{1\max}}\right| = \alpha_1, \left|\dfrac{\partial \Delta h_1}{\partial h_{4\max}}\right| = 0 \\
\left|\dfrac{\partial \Delta h_2}{\partial h_{1\max}}\right| = \dfrac{S_1}{S_2}\alpha_1, \left|\dfrac{\partial \Delta h_2}{\partial h_{4\max}}\right| = 0 \\
\left|\dfrac{\partial \Delta h_3}{\partial h_{1\max}}\right| = 0, \left|\dfrac{\partial \Delta h_3}{\partial h_{4\max}}\right| = 0 \\
\left|\dfrac{\partial \Delta h_4}{\partial h_{1\max}}\right| = 0, \left|\dfrac{\partial \Delta h_4}{\partial h_{4\max}}\right| = \alpha_4
\end{cases}
$$

The results can be used to quantitatively describe the sensitivity of the water surface changes of each lake to the height of the two dams.

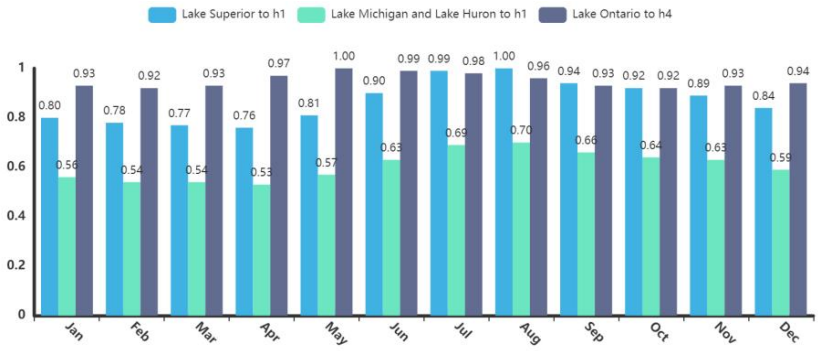

Figure 9: the sensitivity of the water surface changes of each lake to the height of the two dams.

Satisfaction model solution:

Bringing the 2017 Great Lakes water level into the above formula, we get the actual lake level deviation value d'_i. The upper and lower limits of the optimal water level interval solved in the first question are used to solve the ideal deviation values of the four lake levels d_i. Its value is as follows:

Chart 3: True value of d' and Desired value of d'

Lake	True value of d'	Desired value of d'
Lake Superior	0.13	0.24
Lake Michigan and Lake Huron	0.85	0.24
Lake Erie	0.37	0.14
Lake Ontario	0.07	0.14

Through comparison, it is found that the new control measures can better meet the interests of various stakeholders in the vicinity of Lake Superior and Lake Ontario.

3.4 Problem4

3.4.1 Model Establishment

Model assumptions:

The variation of water level in the Great Lakes is mainly affected by three factors: precipitation, ice blockage and evaporation [5].

The Great Lakes have similar climate changes in the same season.

Setting sensitivity index

We set RMSE as an index to measure the sensitivity of water level change. The larger the RMSE value, the more significant the impact of environmental variables on water level, the higher the sensitivity, and the smaller the vice versa. The formula is as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \widehat{x_i}\right)^2}$$

Environmental condition variable:

Suppose that the total precipitation $rain_s$ of the Great Lakes in the same season s changes with the season as follows:

$$rain_s = m_s \cdot S_i, s = 1,2,3,4$$

m_s is Average quarterly precipitation for the Great Lakes .

River discharge during ice blockage period is:

$$Q_s = \alpha_i \cdot M_i \cdot \phi$$

Where ϕ is the ice plugging coefficient, if the ice plugging phenomenon occurs in the Great Lakes, the ice plugging coefficient ϕ is 0, if not, the ice plugging coefficient ϕ is 1.

Evaporation in the Great Lakes $evoporation_s$ is related to temperature T and surface area S_i as follows:

$$evoporation_s = k \cdot T_s \cdot S_i$$

k is 0.7.

Sensitivity analysis model

We establish the relationship between the system performance RMSE and the environment condition variables. Suppose the model is linear:

$$RMSE = a \cdot rain + b \cdot snow + c \cdot evaporation + \varepsilon$$

ε is random error.

Using finite difference method, we estimate the sensitivity of coefficients a,b,c to RMSE. Suppose we take $\delta = 0.01$ meters as the difference distance, constantly changing, and establish the sensitivity equation:

$$\begin{cases} Sensitivity_{rain} = \dfrac{RMSE(a+\delta,b,c) - RMSE(a-\delta,b,c)}{2\delta} \\ Sensitivity_{snow} = \dfrac{RMSE(a,b+\delta,c) - RMSE(a,b-\delta,c)}{2\delta} \\ Sensitivity_{evaporation} = \dfrac{RMSE(a,b,c+\delta) - RMSE(a,b,c-\delta)}{2\delta} \end{cases}$$

3.4.2 Model Solving and Analysis of Results

The relationship model between environmental condition variables and water level was established to control the changes of precipitation, snowmelt and evaporation respectively.

In question 2, we established a control model to predict the change of lake volume. Here, we did not consider the difference between basin area and water area, and established the following equation:

$$\frac{dV_j}{dt} = Vn_j \cdot S_j - \alpha_j \left(h_j - h_{j\max}\right) \cdot S_j = \Delta V_j$$

L_i mainly includes river inflow and environmental impact, and environmental impact mainly includes lake precipitation, ice blockage and evaporation. We give a more specific natural net inflow:

$$Vn_{j,s} = rain_s + Q_s + evoporation_s$$

L_i equals to:

$$Vn_j = m_s \cdot S_j + \alpha_i \cdot M_i \cdot \phi + k \cdot T \cdot S_j$$

We can find the equation of the water level of the lake every quarter:

$$\Delta h_{j,s} = \frac{\Delta V_{j,s}}{S_j} = \frac{Vn_{j,s} - \alpha_{j,s} \cdot h_{j,s} \cdot S_j}{S_j} = \frac{Vn_{j,s}}{S_j} - \alpha_{j,s} \cdot h_{j,s}$$

The sensitivity index formula of lake surface in the same quarter is as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(\Delta h_{j,s}\right)^2}$$

Based on the data collected, we found that Lake Superior, Lake Michigan and Lake Erie have a high probability of ice blockage, especially in January and February. Due to the relatively warm climate of Lake Roentgen and Lake Ontario, the probability of ice blockage is low, so we believe that the ice blockage coefficient ϕ of Lake Superior, Lake Michigan and Lake Erie in spring is 0, and the ice blockage coefficient ϕ of Lake Roentgen and Lake Ontario is 1.

Bring the environmental variable data we collected into the equation calculation.

Finally, by controlling the change of precipitation, the value of sensitivity coefficient a is 4.57

Similarly, the value of b is 2.36 and the value of c is -3.47.

Because the sensitivity coefficients a,b and c of RMSE are large, the control

algorithm is highly sensitive to changes in environmental conditions.

3.5 Problem5

3.5.1 Model Establishment

The fitting function of the annual mean water volume of Lake Ontario $h_{an}(m)$ was established by the least square method. After many attempts, it was found that the significance effect of quadratic fitting was the best:

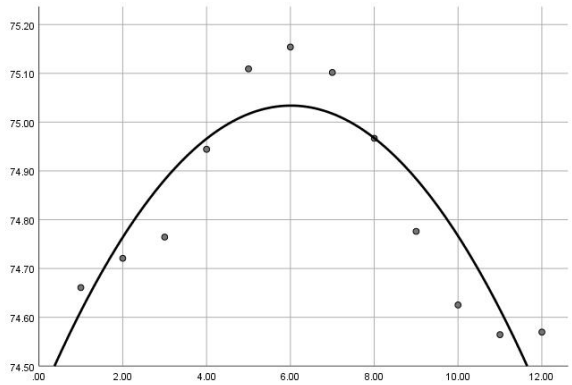

Figure 10：fitting function

The fitted function $h_{on}(m)$ is obtained:

$$h_{on}(m) = -0.017m^2 + 0.202m + 74.428$$

Find the first derivative to get the rate of change of water level $h_{on}'(m)$:

$$h_{on}'(m) = -0.034m + 0.202$$

It shows that according to natural conditions, the water level of Lake Ontario is approximately a quadratic function, the level of the water level can be directly reflected by the function value $h_{on}(m)$, and the fluctuation of the water level can be reflected by the function value $h_{on}'(m)$.

By optimizing the original water level function of Lake Ontario $h_{on}(m)$ to satisfy stakeholders as much as possible, optimization parameters *a,b,c* are added to the optimized water level equation of Lake Ontario:

$$H_{on}(m) = (-0.017 + a)m^2 + (0.202 + b)m + 74.428 + c$$

$$H_{on}'(m) = (-0.034 + 2a)m + 0.202 + b$$

Chart 4:Explanation of Optimization parameter

Demand	Optimization parameter
Water level	a,b,c
Water level fluctuation	a,b

Next, the analytic hierarchy process will be adopted to determine the weight of each stakeholder [6], so as to determine the selection of optimization parameters.

Seasonal expected water level model of Lake Ontario based on Analytic Hierarchy process:

The requirements of the six stakeholders are:

Chart 5:The requirements and weight of the six stakeholder

Stakeholders	Requirement	weight
People who live along Lake Ontario	Medium, stable	w_1
Residents of the Port of Montreal	Low, stable	w_2
Shipping company	High, stable	w_3
fishing-boat	Medium, stable	w_4
environmentalist	Unstable	w_5
Power generation company	High, stable	w_6

Quantify the requirements:

High water level = 1, Low water level = -1, Medium water level = 0

Unstable = 1, Stable = -1

Therefore, the level of the water level is quantified as: $w_3 + w_6 - w_2$ .If the result is greater than 0, the optimization should be biased toward the high water level; if it is less than 0, the optimization should be biased toward the low water level.

The fluctuation of water level can be quantified as follows: $w_5 - w_1 - w_2 - w_3 - w_4 - w_6$. If the result is greater than 0, the optimization should be biased towards the fluctuating water level; if it is less than 0, the optimization should be biased towards the stable water level.

Through these two indexes, the optimization parameters a,b,c can be selected and combined to achieve the effect of optimizing the water level.

It is also stated that a change in water level of 2-3 feet will have a significant impact on stakeholders, and from a stability point of view, this optimization will be controlled within 2 feet (0.6m) :

$$H_{on}'(m) - H_{on}(m) < 0.6, m = 1, 2, \cdots, 12$$

3.5.2 Model Solving and Analysis of Results

Through analytic hierarchy process, it is calculated that:

$$w_p = [0.3109, 0.3109, 0.0552, 0.0528, 0.1243, 0.1459]$$

It is calculated that:

The quantified value of the fluctuation of the water level $w_3 + w_6 - w_2$ is -0.1097, indicating that the optimization should be biased towards the low water level.

The quantified value of the fluctuation of the water level $w_5 - w_1 - w_2 - w_3 - w_4 - w_6$ is -0.7514, indicating that the optimization should be biased towards the stable water level.

Select optimization parameters:

$$a = 0.0045, b = -0.052, c = -0.128$$

Substitute the optimization parameters into the optimization equation:

$$H_{on}(m) = -0.0125m^2 + 0.1500m + 74.3000$$

The water level function image after optimization within one year is compared with the function image before optimization.

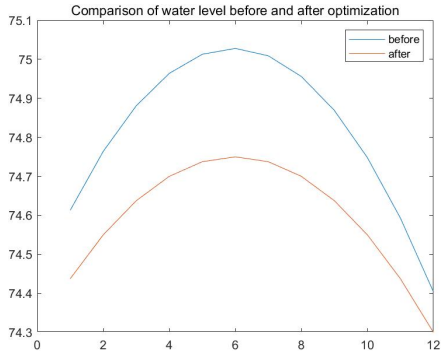

Figure 11:Comparison of water level before and after optimization

As can be seen from the figure, the overall water level is reduced and the fluctuation of water level is reduced, achieving the goal of optimizing water level and taking into account the needs of various stakeholders.

4 Conclusion

This paper mainly studies the water level regulation mechanism of the Great Lakes, so we need to build a mathematical model to provide a management plan for the water level regulation of the Great Lakes. The differential equation of lake system and the flow rate model of river are established, and the root-mean-square error of water level control is selected for sensitivity analysis. This will mathematically analyze the rivers and lakes in the Great Lakes basin and enable us to complete the program we need more systematically and scientifically. In fact, in the research process of this paper, it is difficult to quantify relevant issues due to insufficient data. We also ignored the influence of small lakes and rivers such as Lake.S.T.Clair on the results, and idealized it to be attributed to large lakes and rivers. In addition, the effects of climate conditions on water levels in the Great Lakes region are not carefully analyzed. In future studies, when formulating plans for the water level regulation of the Great Lakes, the above conditions can be more widely queried for quantification, so as to make the model more consistent with the actual water level change of the Great Lakes. Finally, I would like to thank the Organizing Committee for giving us such an opportunity and hope that our solution can be helpful to the water level management of the Great Lakes.

5 Reference

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Memo

Caption： Overview of the Great Lakes water level regulation control scheme

Date and time： 5 February 2024 at 14:00 BST

To： IJC

Introductory：

As the largest group of freshwater lakes in the world, the Great Lakes of North America cover shipping, irrigation, residents, businesses, as well as local economic development and ecological environment. The main purpose of this paper is to establish a control mechanism to manage the water level of the Great Lakes, balance the flood disaster caused by less drainage and evaporation and the impact of excessive displacement restricting the passage of large ships on local economic development. We will introduce you to the solution we have specified for each problem and explain the features and advantages of our solution compared to the original solution.

Main text：

First, through the analysis of the water level data of the Great Lakes from 2000 to 2022, the annual comprehensive average water level is obtained. We used 2 feet above the equilibrium water level to get the floating rate of water level. According to the floating rate of water level, we can judge that the annual comprehensive average water level we got is in the best range of water level, which meets the requirements of stakeholders.

Secondly, we simulate the lake system and construct a river flow rate model. According to the principle of long-term climate stability in the Great Lakes basin, the assumption of full river flow is put forward. This problem also needs to control the water quantity by adjusting the height of the two DAMS. We set up a differential equation for the flow of the Great Lakes, which will more accurately obtain the optimal water level of the Great Lakes.

Then, in the sensitivity analysis of the two DAMS, we mainly analyzed the water level data in 2017. By solving the derivative function of the above differential equation, the sensitivity of the dam when adjusting the water level is obtained. We choose the root-mean-square error (RMSE) of water level control as the sensitivity index, and establish the variable relationship between system performance RMSE and environmental conditions. After our calculations, we believe that the new measures will enable stakeholders to provide better feedback on the actual water level.

Finally, in our extensive analysis of Lake Ontario, we focused on stakeholders' conflicts of interest due to high and low water levels.

Conclusions and recommendations：

These are our general plans for the regulation and management of water levels in the Great Lakes. We use a variety of mathematical calculations to reflect the correlation of factors more accurately than geographical surveys. It is hoped that our scheme can help to regulate and manage the water level of the Great Lakes, so that the Great Lakes can play a better role.